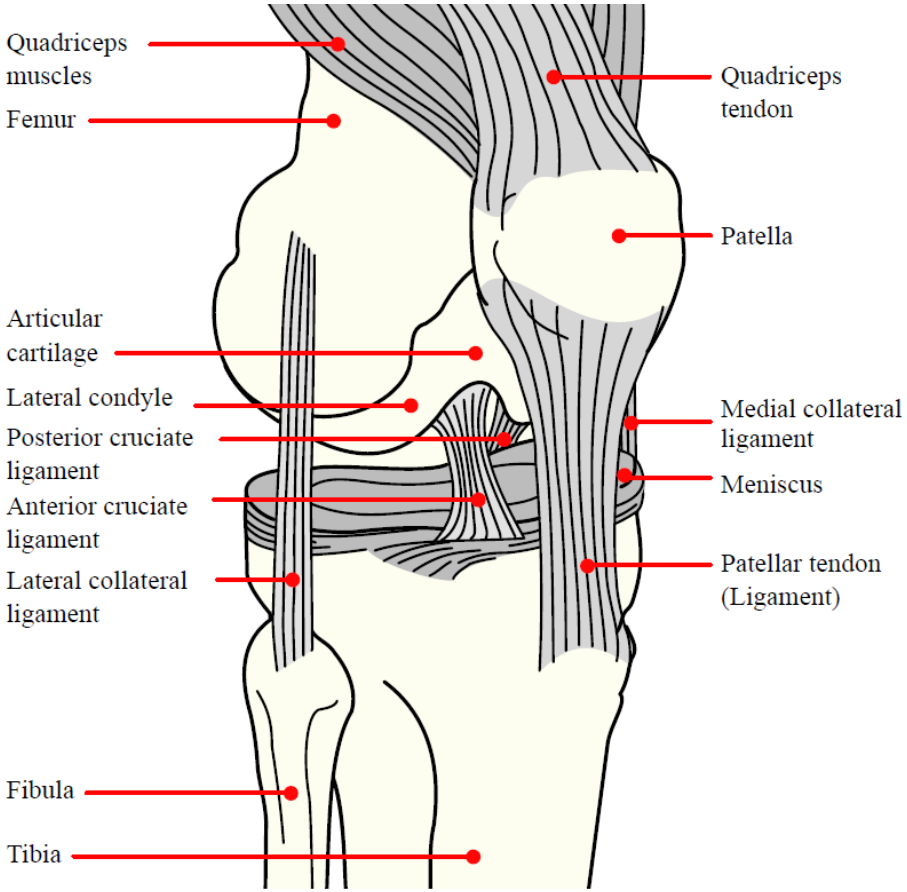
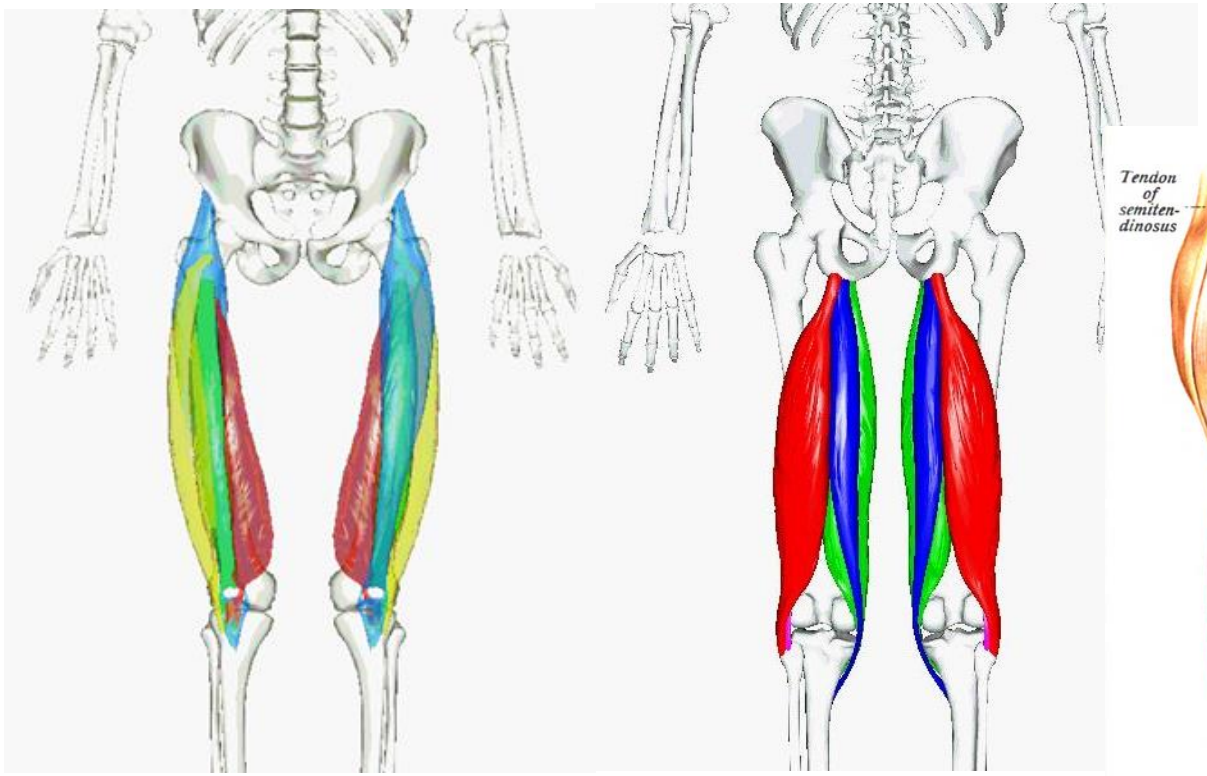
Anatomy – knee



By Mysid - Vectorized and colorized in Inkscape, based on Image:Knee diagram.png., Public Domain, https://commons.wikimedia.org/w/index.php?curid=4004042

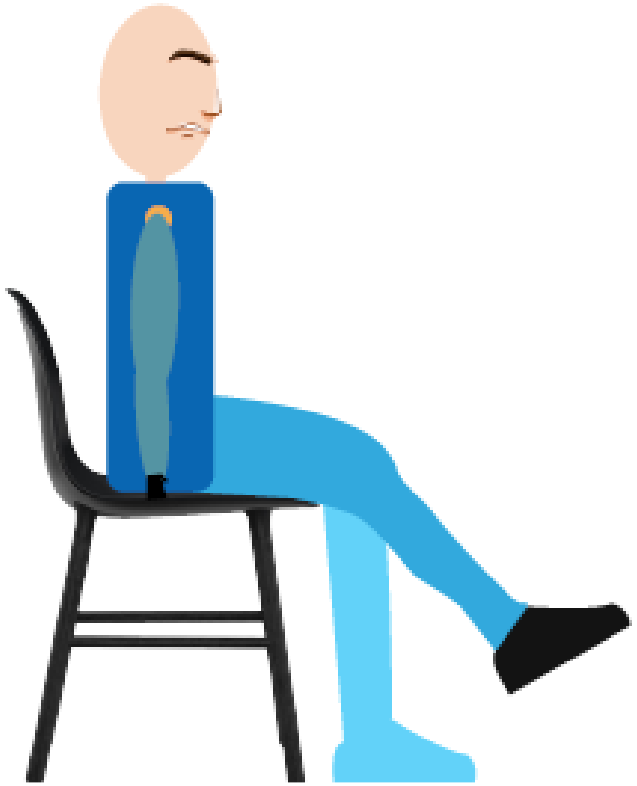


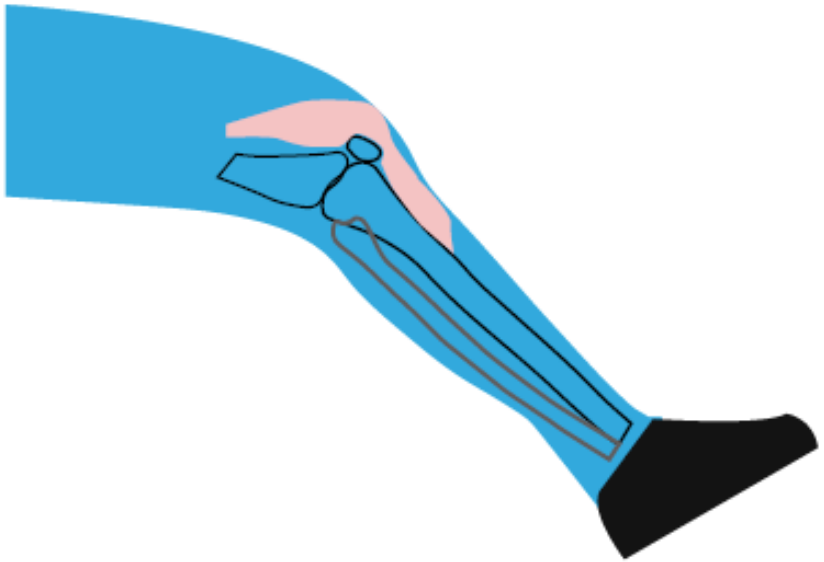
By BodyParts3D/Anatomography - BodyParts3D/Anatomography, CC BY-SA 2.1 jp, https://commons.wikimedia.org/w/index.php?curid=33808194



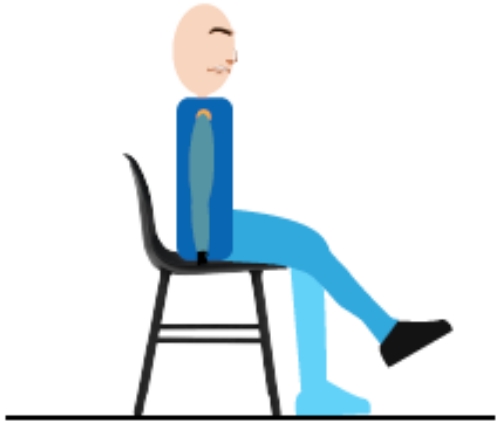
By Dr. Johannes Sobotta - Sobotta's Atlas and Text-book of Human Anatomy 1909, Public Domain, https://commons.wikimedia.org/w/index.php?curid=29822860

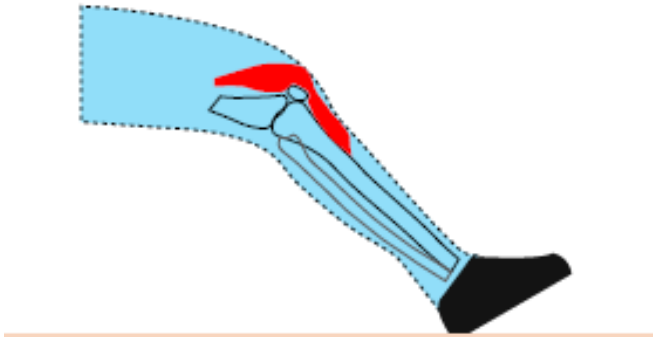
Static analysis - knee

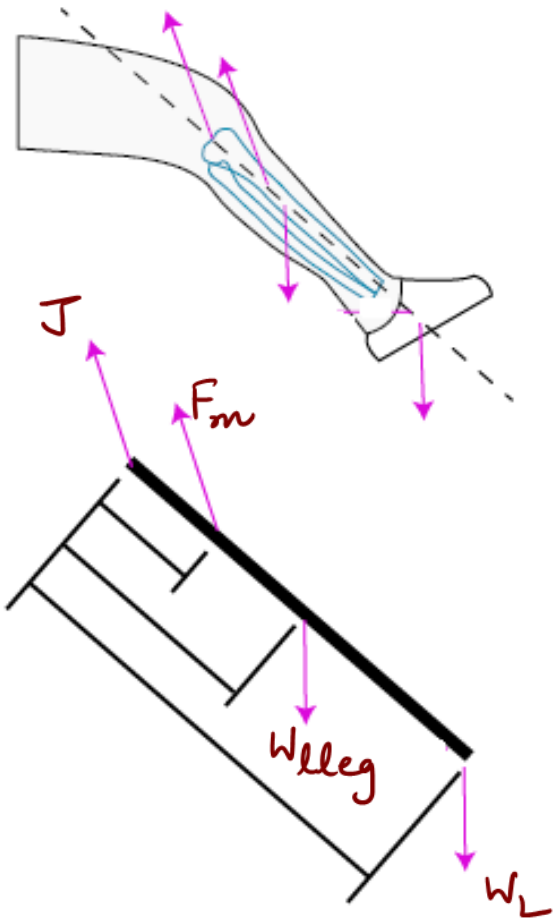




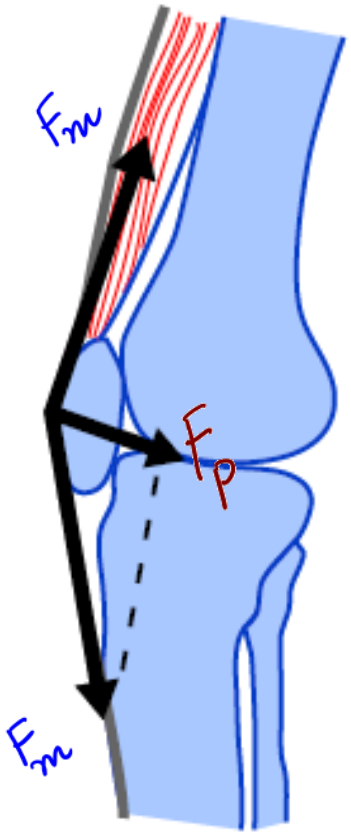
Static analysis - knee

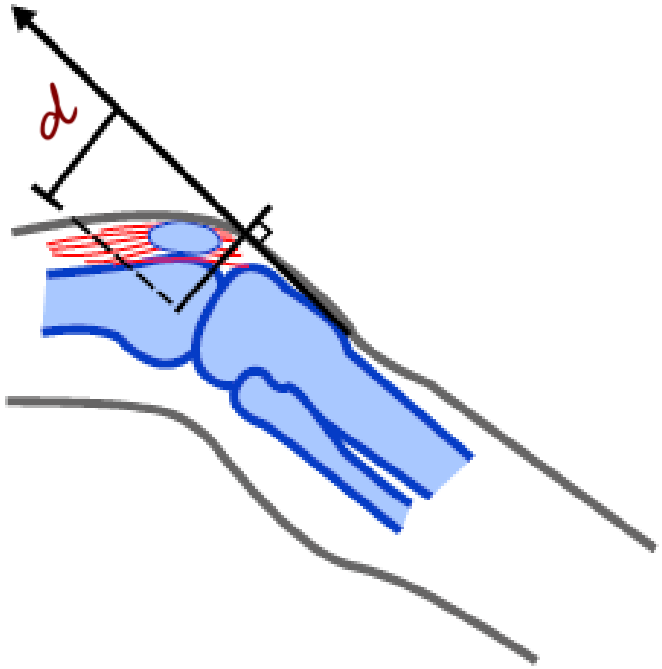




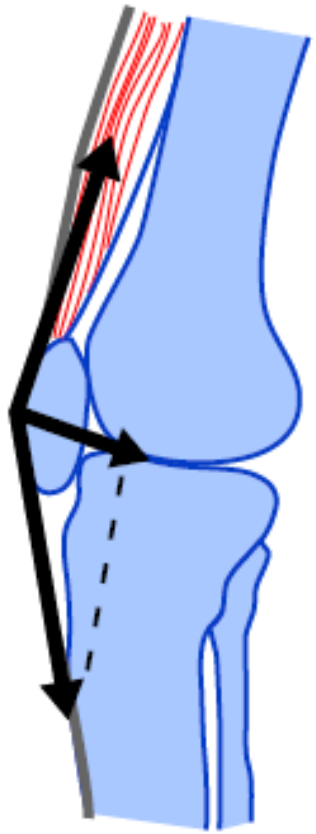


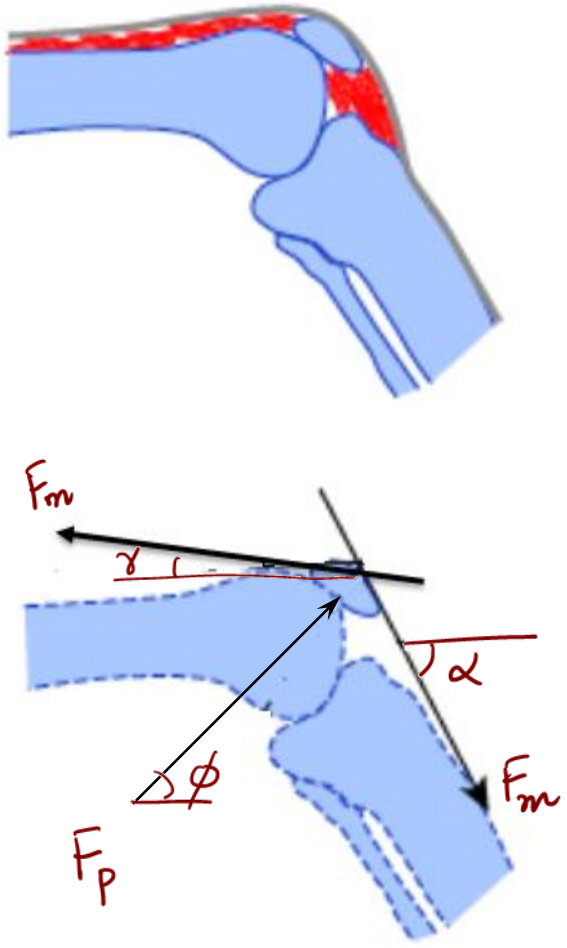
Role of patella





Static analysis - knee





Patellofemoral joint

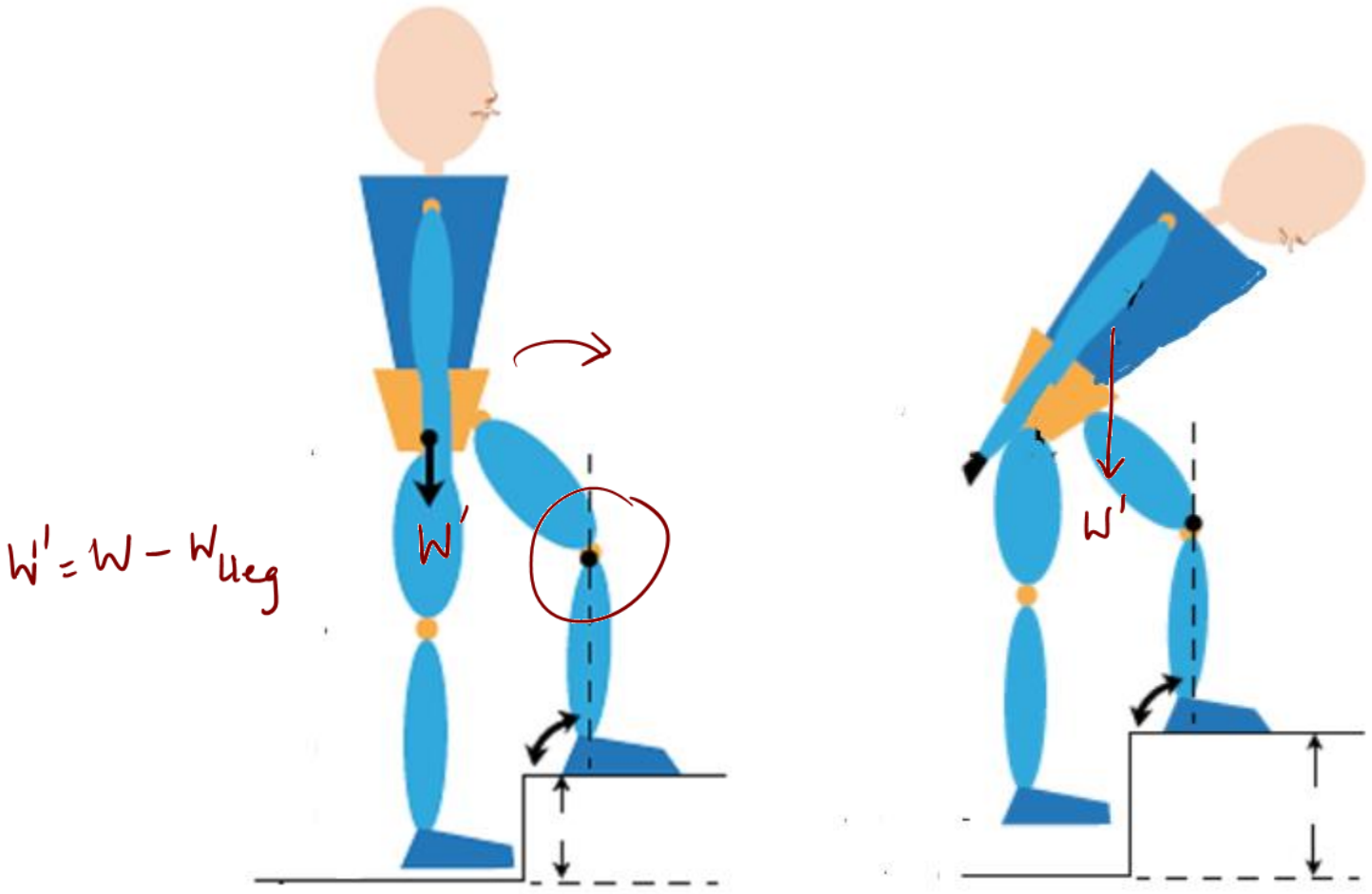
$$-F_m \cos\gamma + F_m \cos\alpha + F_p \cos\phi = 0$$

$$F_m \sin\gamma - F_m \sin\alpha + F_p \sin\phi = 0$$

$$F_p = \left(\frac{\cos\gamma - \cos\alpha}{\cos\phi}\right) F_m$$

$$\phi = \tan^{-1}\left(\frac{\sin\alpha - \sin\gamma}{\cos\gamma - \cos\alpha}\right)$$

Step climbing



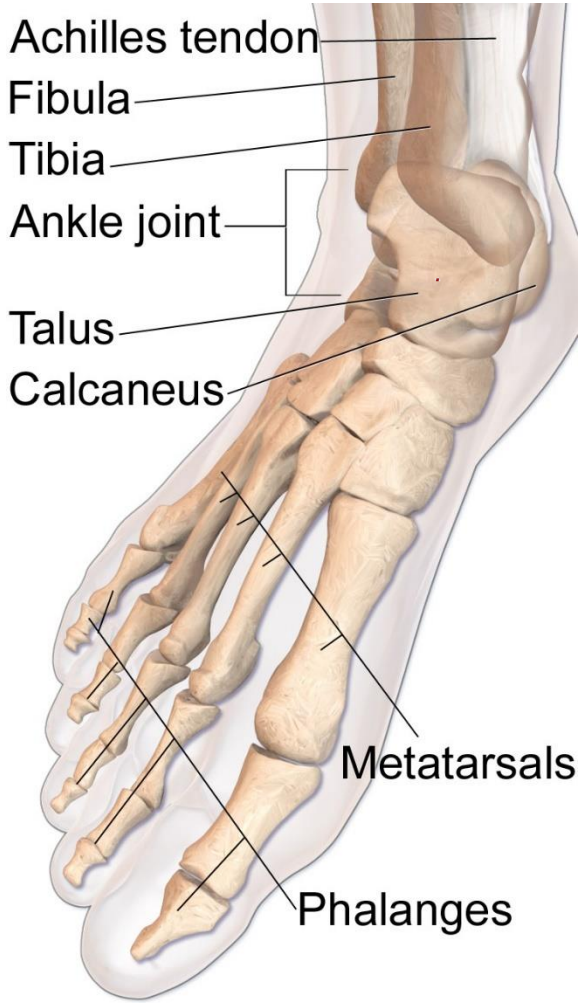
Anatomy – ankle

Tibia + fibula form a joint with talus

Functions like a hinge joint

Dorsiflexion & Plantarflexion of the foot

Tibiotalar joint

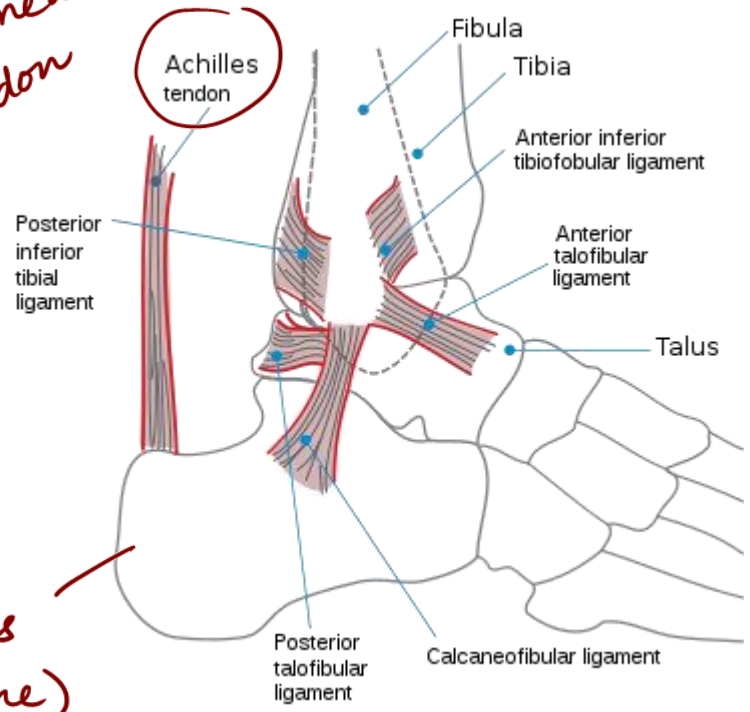


Lower Leg and Foot

Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). DOI:10.15347/wjm/2014.010. ISSN 2002-4436. - Own work, CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=29849183

Anatomy – ankle

Calcaneal tendon

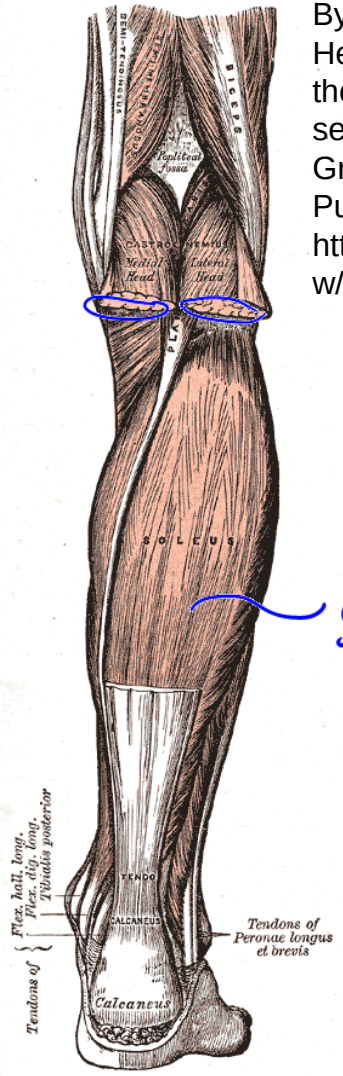


Calcaneus (Heel bone)

By An svg version of image:Ankle.PNG by me (Jak). - A vector version of the original image found here [1], Public Domain, https://commons.wikimedia.org/w/index.php?curid=1981086

Gastroc also knee flexor

Superficial – gastroc
Deep – soleus



By Henry Vandyke Carter - Henry Gray (1918) Anatomy of the Human Body (See "Book" section below)Bartleby.com: Gray's Anatomy, Plate 438, Public Domain, https://commons.wikimedia.org/w/index.php?curid=529080

Plantarflexion

Gastrocnemius

+

Soleus

Calf muscles
Triceps surae

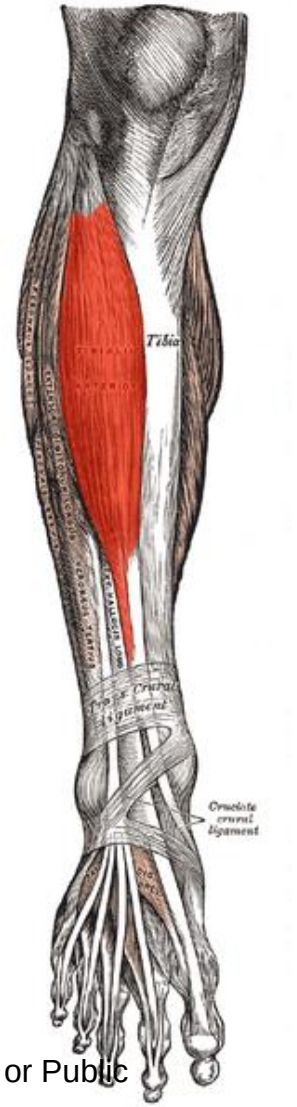


By Dr. Johannes Sobotta - Sobotta's Atlas and Text-book of Human Anatomy 1909, Public Domain, https://commons.wikimedia.org/w/index.php?curid=29822860

Anatomy – ankle

Tibialis anterior

Dorsiflexion of the foot

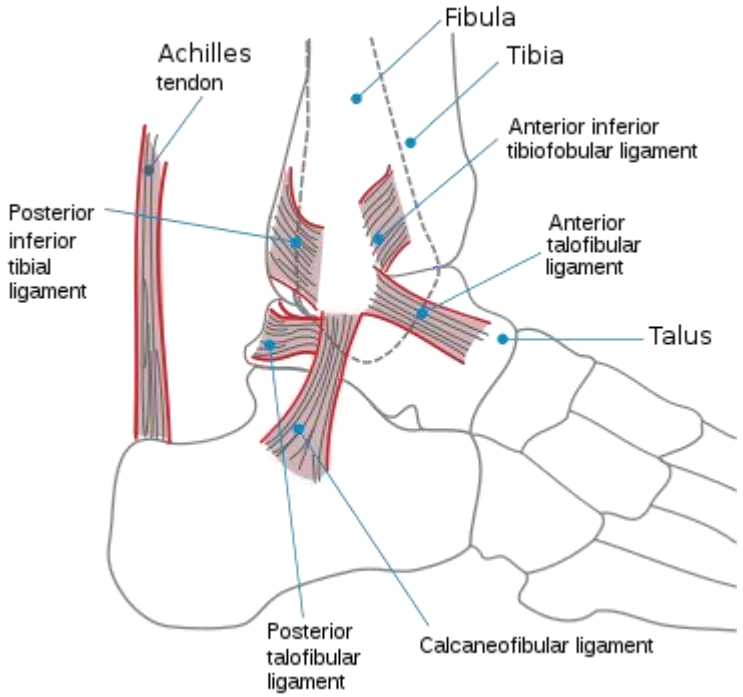


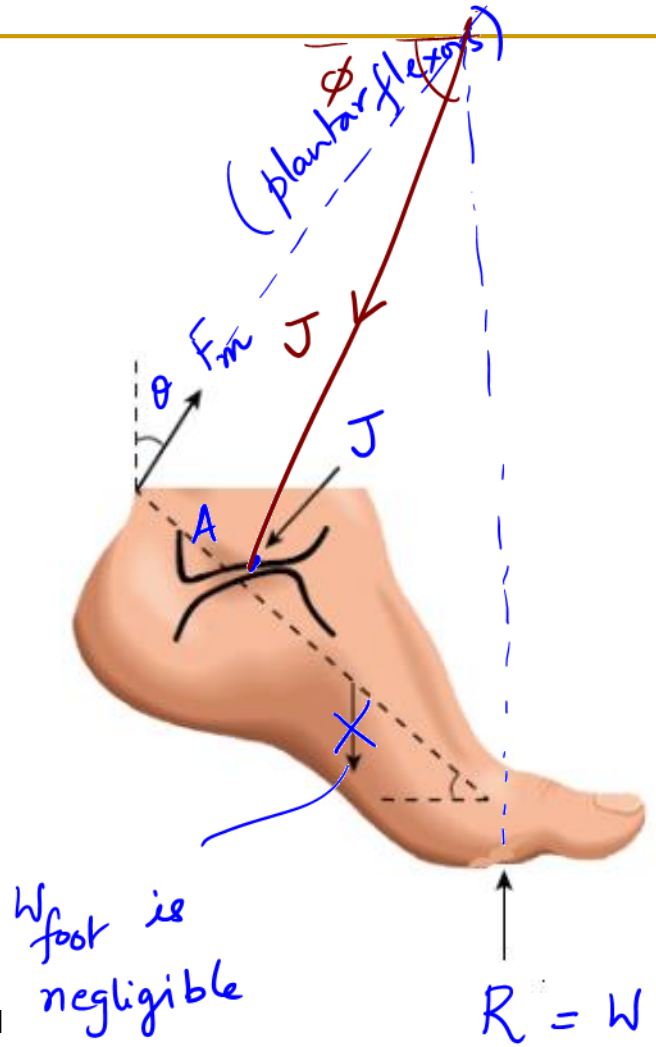
By modified by Uwe Gille (Gray437.png) [Public domain, Public domain or Public domain], via Wikimedia Commons



By Dr. Johannes Sobotta - Sobotta's Atlas and Text-book of Human Anatomy 1909, Public Domain, https://commons.wikimedia.org/w/index.php?curid=29822860

Mechanics of the ankle





Mortise & tenon joint



J - magnitude unknown & direction unknown

By An svg version of image:Ankle.PNG by me (Jak). - A vector version of the original image found here [1], Public Domain, https://commons.wikimedia.org/w/index.php?curid=1981086

Kinetics: linear motion

- Newton's second law
  - Conservation of linear momentum
- Newton's third law

Any object in space can be idealized as a system of particles

For each particle i in the system, Newton's laws of motion will hold

$$\bar{F}_i + \sum_{j=1}^{n} \bar{f}_{ij} = m_i \bar{a}_i$$

$\bar{F}_i$ = net force exerted on particle i by particles external to the system

$\bar{f}_{ij}$ = force exerted by particle j on particle i

A particle cannot exert a force on itself $\therefore \bar{f}_{ii} = 0$

Summing over all the particles in the system

$$\sum \bar{F}_i + \sum\sum \bar{f}_{ij} = \sum m_i \bar{a}_i$$

$\sum \bar{F}_i$ — external forces on the system : contact forces, gravitational forces

$\sum\sum \bar{f}_{ij}$ — sum of all the internal forces in the system

$= 0$ $\quad f_{ij} = -f_{ji}$ Newton's third law

$\therefore$ Internal forces don't contribute to the acceleration of the system

$$\therefore \sum \bar{F}_i = \sum m_i \bar{a}_i$$

Now, the linear momentum $\bar{L}$ of a system of particles is defined as $\bar{L} = \sum m_i \bar{v}_i$

$$\frac{d\bar{L}}{dt} = \sum m_i \frac{d\bar{v}_i}{dt} = \sum m_i \bar{a}_i$$

$$\therefore \sum \bar{F}_i = \frac{d\bar{L}}{dt}$$

$\sum \bar{F}_i = 0$ statics

(CoM)

Centre of mass is defined by

$$(\sum m_i)\,\bar{r}_c = \sum m_i r_i$$

$\sum m_i$ is the total mass in the system

$\bar{r}_c$ is the position vector of the CoM

m_i is the mass of the ith particle

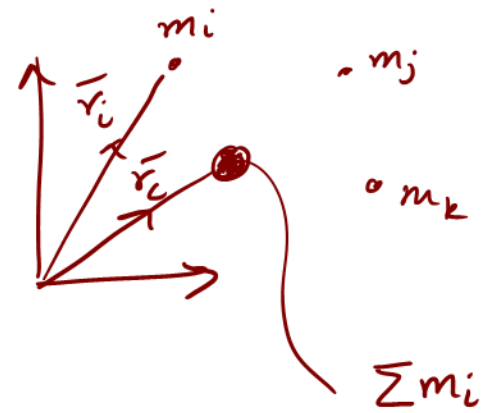
r_i is the position vector of the ith particle

Velocity of the CoM $\quad \bar{v}_c = \dfrac{d\bar{r}_c}{dt}$

Acceleration of the CoM $\quad \bar{a}_c = \dfrac{d\bar{v}_c}{dt}$

If $\quad \bar{r}_c = \dfrac{\sum m_i r_i}{\sum m_i}$, then $\quad \bar{v}_c = \dfrac{d\bar{r}_c}{dt} = \dfrac{\sum m_i \bar{v}_i}{\sum m_i}$

$$\bar{a}_c = \frac{\sum m_i \bar{a}_i}{\sum m_i}$$



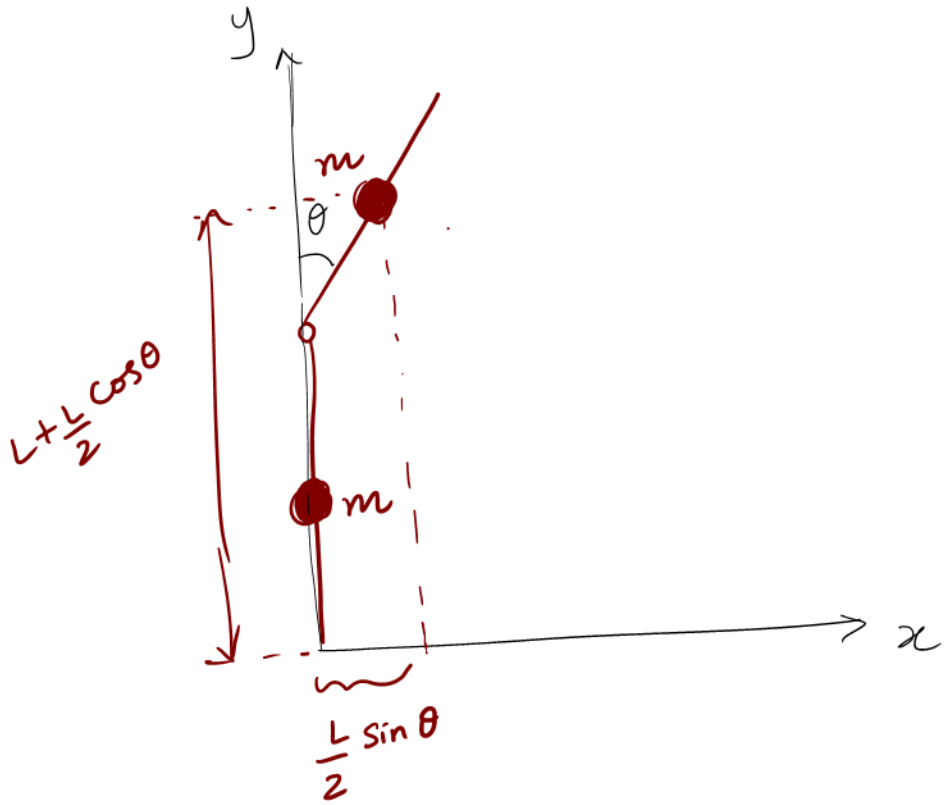
$$\sum \bar{F}_i = \sum m_i a_i$$

$$= (\sum m_i)\,\bar{a}_c$$

∴ The net external force acting on a system of particles is equal to the mass of the system times the acceleration of its CoM

Say, I represent the human body as two rods, each of length L & mass m

What is the CoM of this system?



$$(m+m)\,x_c = m(0) + m\frac{L}{2}\sin\theta$$

$$x_c = \frac{L}{4}\sin\theta$$

$$(m+m)\,y_c = m\frac{L}{2} + m\left(L+\frac{L}{2}\cos\theta\right)$$

$$y_c = \frac{3L}{4} + \frac{L}{4}\cos\theta$$



y_c = 55% of height for females
57% of height for males
] normal standing position